Technoverse Hackathon

2026

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<Smart Automatic Timetable Generation> | Business Plan (1/2)

WHY

Explain the Problem

Problem Description & Business Scenario: Academic institutions face a combinatorial explosion in timetable scheduling due to multiple interdependent constraints like faculty availability, classrooms, and subjects. This results in frequent scheduling conflicts, errors, and significant administrative inefficiency.
Problem Scope: The problem involves handling complex coordination of student batches, faculty schedules, and physical resources under multiple constraints. These dependencies make it difficult to generate a conflict-free and efficient timetable manually.
Target Users/Stakeholders: University Administrators, Faculty Members, Students, and Department Heads.

How

Explain the Solve

Solution Overview: The system provides an automated timetable generation engine using a priority-based greedy algorithm to create conflict-free schedules. It includes key features like faculty and classroom allocation, lab scheduling, elective slot grouping, and real-time clash detection. Additionally, it offers a manual editing interface, interactive timetable view, and real-time slot availability feedback for better administrative control.
Technical Details: Built using the MERN stack, the system employs a custom greedy scheduler with efficient clash detection. It uses a 2D timetable structure and global tracking of resources to maintain accuracy and consistency.
Innovation: The project introduces slot grouping for elective scheduling and a prompt-based timetable generation system. It ensures zero-conflict schedules with real-time clash detection and feedback.
Market Potential: The system has strong potential across schools, colleges, and training institutes, with a scalable architecture suitable for SaaS-based deployment in large-scale educational environments.

WHAT

Value proposition

Primary benefits: Ensures no clashes in faculty and classroom scheduling, supports multiple subjects and electives at the same time, and gives real-time control to administrators.
Efficiency and flexibility: Provides a fast one-click timetable generator, supports both Excel data and text input, and allows easy manual editing.
Time and cost saving: Reduces manual work by up to 90% and improves classroom and lab usage through smart scheduling.
Scalability : Can grow from a single department to multiple campuses, handling large-scale scheduling easily.
Social Impact: Improves education quality by reducing teacher workload and ensuring smooth, clash-free schedules for students.

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Financials & Timelines | Business Plan (2/2)

Investments

What does it take & How much does it cost to solve?

Cloud: Requires cloud infrastructure and database setup with low to moderate monthly cost.
Process Re-engg: Needs data structuting and initial setup effort with moderate one time cost.
Modernization: Involves system development using MERN stack with minimal software cost..
AI: Uses AI to process text inputs and generate intelligent timetables automatically with usage based cost
Tools: Requires deployment and monitoring tools with low or high cost options.

Returns

Quantify the benefits & What if I don't solve?

ROI / Revenue Growth: Increases resource usage and allows more students and courses without extra infrastructure.
Happy Customer: Provides clash-free schedules for smooth student learning and stable faculty plans.
Margin Improvement: Reduces manual work and costs while maximizing resource efficiency.
Employee Experience: Lowers admin stress and ensures balanced workload for faculty.

Timelines

Time to realize benefits

Time to solve: The system can be fully developed within **2–4 weeks**, including setup, data structuring, and testing.
Milestones:
1. Data collection and structuring of faculty, rooms, and subjects.
2. Development of scheduling engine and core system features.
3. Integration of clash detection, slot grouping, and UI interface.
4. Testing, validation, and deployment of the system.
5. Final optimization and real-time usage by administrators.

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Thank you

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